37. Let $\{N(t), t \geq 0\}$ be a Poisson process with rate λ. For $i \leq n$ and $s < t$,

(a) find $P(N(t) = n | N(s) = i)$;

(b) find $P(N(s) = i | N(t) = n)$.

55. Consider a single server queuing system where customers arrive according to a Poisson process with rate λ, service times are exponential with rate μ, and customers are served in the order of their arrival. Suppose that a customer arrives and finds $n - 1$ others in the system. Let X denote the number in the system at the moment that customer departs. Find the probability mass function of X.

59. Cars pass an intersection according to a Poisson process with rate λ. There are 4 types of cars, and each passing car is, independently, type i with probability p_i, $\sum_{i=1}^{4} p_i = 1$.

(a) Find the probability that at least one of each of car types 1, 2, 3 but none of type 4 have passed by time t.

(b) Given that exactly 6 cars of type 1 or 2 passed by time t, find the probability that 4 of them were type 1.

68. Suppose that electrical shocks having random amplitudes occur at times distributed according to a Poisson process $\{N(t), t \geq 0\}$ with rate λ. Suppose that the amplitudes of the successive shocks are independent both of other amplitudes and of the arrival times of shocks, and also that the amplitudes have distribution F with mean μ. Suppose also that the amplitude of a shock decreases with time at an exponential rate α, meaning that an initial amplitude A will have value $Ae^{-\alpha x}$ after an additional time x has elapsed. Let $A(t)$ denote the sum of all amplitudes at time t. That is,

$$A(t) = \sum_{i=1}^{N(t)} A_i e^{-\alpha(t - S_i)}$$

where A_i and S_i are the initial amplitude and the arrival time of shock i.

(a) Find $E[A(t)]$ by conditioning on $N(t)$.

(b) Without any computations, explain why $A(t)$ has the same distribution as does $D(t)$ of Example 5.21.

63. Consider an infinite server queuing system in which customers arrive in accordance with a Poisson process with rate λ, and where the service distribution is exponential with rate μ. Let $X(t)$ denote the number of customers in the system at time t. Find

(a) $E[X(t+s)|X(s)=n]$;

(b) $\text{Var}(X(t+s)|X(s)=n)$.

Hint: Divide the customers in the system at time $t+s$ into two groups, one consisting of "old" customers and the other of "new" customers.

(c) If there is currently a single customer in the system, find the probability that the system becomes empty when that customer departs.